Predicting fire **spread** and behaviour on the fireline. **Wildfire** analyst pocket: A mobile app for **wildland fire prediction**

S Monedero, J Ramirez, A Cardil - Ecological Modelling, 2019 - Elsevier

… at providing real-time analysis of **wildfire** behaviour and **spread** in order to support real-… of **spread**, flame length, fireline intensity, crown fire potential, among others) for each **simulation** …

☆ Save 99 Cite Cited by 72 Related articles All 6 versions

Assessing WildfireGPT: a comparative analysis of AI models for quantitative **wildfire spread prediction**

M Ramesh, Z Sun, Y Li, L Zhang, SK Annam, H Fang… - Natural Hazards, 2025 - Springer

… for **wildfire** forecasting, focusing on its limitations in quantitative predicting Fire Radiative Power (FRP) **spread** and comparing its performance with a specialized predictive **model** based …

☆ Save 99 Cite Cited by 2 Related articles All 3 versions

Developing the US **wildland fire** decision support system

EK Noonan-Wright, TS Opperman… - Journal of …, 2011 - Wiley Online Library

… WFDSS accesses national weather data and forecasts, fire **behavior prediction**, economic … The objective of this paper is to **review** the US **Wildland Fire** Decision Support System. We …

☆ Save 99 Cite Cited by 179 Related articles All 8 versions

Global pyrogeography: the current and future distribution of **wildfire**

MA Krawchuk, MA Moritz, MA Parisien, J Van Dorn… - PloS one, 2009 - journals.plos.org

… fire in research on global vegetation-climate change **dynamics** and conservation planning. … we built our **wildfire** occurrence models from the GFDL CM2.1 historical **simulation** data, from …

☆ Save 99 Cite Cited by 1133 Related articles All 25 versions »

Review of formal methodologies for wind–slope correction of **wildfire** rate of **spread**

JJ Sharples - International Journal of **Wildland Fire**, 2008 - CSIRO Publishing

… to compare the quality of **prediction** of various fire **spread** models incorporating wind and … a detailed **review** of the formal wind–slope correction methods used in a suite of fire **spread** …

☆ Save 99 Cite Cited by 82 Related articles All 11 versions

Wildfirespreadts: A dataset of multi-modal time series for **wildfire spread prediction**

S Gerard, Y Zhao, J Sullivan - Advances in Neural …, 2023 - proceedings.neurips.cc

… of a **wildfire**. The only dataset we are aware of that represents **wildfire spread** predictions at a … In section 3, we describe related work in the field of **wildfire spread prediction**. We apply …

☆ Save 99 Cite Cited by 25 Related articles All 4 versions »

Generative AI for Predicting 2D and 3D **Wildfire Spread**: Beyond Physics-Based Models and Traditional Deep Learning

H Xu, S Zlatanova, R Liang, I Canbulat - arXiv preprint arXiv:2506.02485, 2025 - arxiv.org

… apply generative AI models to **wildfire spread prediction**, discussing the underlying rationale … **review** of studies applying generative AI to **wildfire prediction**, highlighting their strengths for …

☆ Save 99 Cite Related articles All 2 versions »

Prediction of **forest fire spread** rate using UAV images and an LSTM **model** considering the interaction between fire and wind

X Li, H Gao, M Zhang, S Zhang, Z Gao, J Liu, S Sun… - Remote Sensing, 2021 - mdpi.com

… new neural network **model** to predict the **spread** rate of the **forest fire**. In … **forest fire spread** speed can be predicted more accurately if the wind speed is considered in the **prediction model**…

☆ Save 99 Cite Cited by 34 Related articles All 8 versions »

FireCast: Leveraging Deep Learning to Predict **Wildfire Spread**.

D Radke, A Hessler, D Ellsworth - IJCAI, 2019 - ijcai.org

… areas surrounding a burning **wildfire** have high-risk of near-future **wildfire spread**, based on historical … to a random **prediction model** and a commonly used **wildfire spread model**, Farsite, …

☆ Save 99 Cite Cited by 159 Related articles All 4 versions »

Tamm **review**: The effects of prescribed fire on **wildfire** regimes and impacts: A framework for comparison

ME Hunter, MD Robles - Forest Ecology and Management, 2020 - Elsevier

… **wildfire** regimes, and their associated ecological and societal effects, focusing specifically on **simulation** modeling studies that consider **wildfire** … ecosystem carbon **dynamics** are much …

☆ Save 99 Cite Cited by 105 Related articles All 10 versions